SANJYOT DAHALE

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Lead Unreal Engine Developer and CTO with 5+ years of experience specializing in real-time systems, XR development, and high-performance rendering pipelines.

WORK EXPERIENCE

LIMINAL XR SOLUTIONS

CTO / Lead Unreal Engine Developer | Jan 2023 - Current

- Managed a cross-functional team of 7 through the full SDLC, establishing Perforce version control and technical standards that led to the successful publishing of *Riddler's Ransom* on the Meta Quest Store.
- Developed custom C++/Blueprint modules, including a multiplayer server-client framework and WebSocket system for real-time MR data synchronization.
- Architected a high-bandwidth stadium-to-CDN 360° VR broadcast pipeline for IPL 2023, managing real-time video ingestion, encoding, and global delivery for live audiences.
- Co-authored a published technical paper in the **SMPTE Motion Imaging Journal** and currently spearheading R&D into local LLM integration (FunctionGemma) for AI-driven interactivity in Unreal Engine applications.

LIMINAL XR SOLUTIONS

Unreal Engine Developer / VP Supervisor | Aug 2020 - Jan 2023

- Researched and implemented end-to-end Virtual Production workflows (ICVFX, Green-screen, Previz) in Unreal Engine, establishing the company's foundational real-time rendering standards.
- Integrated complex hardware ecosystems into Unreal Engine, including stYpe/Mo-Sys camera tracking, XSens/Qualisys Motion Capture, and Blackmagic Design broadcast hardware.
- Defined graphical limitations and performance budgets for asset teams, performing optimization passes on lighting, shaders, and geometry to maintain real-time broadcast and production frame rates.
- Spearheaded the hiring and technical onboarding of Unreal Engine developers and artists, designing training modules for advanced engine workflows and real-time best practices.

PROJECTS

RIDDLER'S RANSOM | META QUEST 3 (UNREAL ENGINE)

- Spearheaded the technical migration of a PC-VR game jam prototype to the Meta Quest 3 standalone platform using the Meta-fork of Unreal Engine.
- Improved performance from **40 FPS to a stable 72 FPS** through rigorous creative asset optimization and technical profiling.
- Managed the end-to-end packaging and submission process, ensuring all technical and performance requirements for the Meta Quest Store were met.

MIXED REALITY CRICKET ANALYTICS FOR BROADCAST | STAR SPORTS (UNREAL ENGINE)

- Architected a custom multiplayer server-client system to synchronize Mixed Reality (HoloLens 2) sessions with Virtual Production clients for real-time broadcast integration.
- Led a massive Blueprint-to-C++ port of the core application logic, resulting in a **70% increase in performance** for complex simulation scenarios.
- Co-authored a peer-reviewed technical paper in the **SMPTE Motion Imaging Journal** detailing the system's novel integration of MR and live broadcast pipelines.

MONTRA ELECTRIC RACING | PC (UNREAL ENGINE)

- Integrated Logitech G29 SDK for immersive haptic feedback and engineered an administrative server-client module to synchronize race states across local clients.
- Helped refine vehicle physics and level design through iterative testing to ensure high-quality gameplay for a high-traffic exhibition environment.

REFRAME 360 - A VR TO 2D WARPING TOOL | R&D PROJECT (UNREAL ENGINE)

- Engineered a low-latency video pipeline in Unreal Engine that uses **custom HLSL shaders** to transform live 4K equirectangular VR feeds into standard 2D broadcast perspectives, such as 'Tiny Planet' or wide-angle views
- Created an operator preset and UI system using Blueprints and UMG to allow real-time color grading and camera manipulation via game controllers.

SKILLS

Languages
C++
Unreal Blueprints
GDScript
HLSL
Python

Engines/Tools
Unreal Engine 4/5
Godot 4
OpenGL
Visual Studio
Perforce
Git

Hardware
Motion Capture (Qualysis, XSens)
Camera Tracking (stYpe, Mo-Sys)

PUBLICATIONS

IMMERSIVE AND INTERACTIVE AR GRAPHICS AND ENVIRONMENTS FOR BROADCAST APPLICATIONS

SMPTE Motion Imaging Journal | Jul 2024

DOI: 10.5594/JMI.2024/IHPC9661

EDUCATION

UNIVERSITY OF MUMBAI

Bachelor of Engineering: Computer Engineering | Jan 2016 - 2020

CERTIFICATIONS

COMPUTER GRAPHICS WITH MODERN OPENGL AND C++

Udemy | Jan 2025

UNREAL ENGINE C++ DEVELOPER: LEARN C++ AND MAKE VIDEO GAMES

Udemy | Aug 2019

LANGUAGES

English
Fluent

Marathi
Native

Hindi
Native